Type I

Persons handling refrigerant during the maintenance, service, or repair of small appliances must be certified as either a Type I Technician or as a Universal Technician, with the exception of motor vehicle air conditioning (MVAC).

The EPA definition of a small appliance includes; products manufactured, fully charged, and hermetically sealed in a factory containing five pounds or less of refrigerant. Split-systems may not be serviced by Type I technicians. The sale of CFC, HCFC, HFC, and HFO refrigerants to service or install refrigeration and air-conditioning equipment is restricted to technicians who are EPA certified in refrigerant recovery.

MVAC-like systems do not meet the criteria for Type I appliance certification no matter the type or quantity of refrigerant. HCFCs and HFCs such as R-404A cannot be used for new Type I appliances.

When retrofitting or converting a system to use a different refrigerant, only EPA-approved substitutes can be used. Remember, there is no such thing as a “drop-in” substitute; EPA does not approve any substitute refrigerants as a direct drop-in replacement to service an R-22 or other system.

Recovery Equipment

Recovery equipment used during the maintenance, service, or repair of small appliances that use CFCs, HCFCs, and HFCs must be certified by an EPA-approved laboratory.

Persons who open or perform maintenance, service, or repair of Type I appliances containing a CFC, HCFC, or HFC, can use either self-contained (active) or passive recovery equipment. A domestic refrigerator is an example of an appliance on which it would be permissible to use a passive recovery device. System-dependent or passive recovery equipment use for an appliance containing HCFCs or HFCs is limited to a maximum charge of 15 pounds.

Maintenance practices for recovery equipment should include checking for refrigerant leaks and oil levels on a regular basis. A refrigerant scale or tank internal float device are acceptable methods for monitoring that the fill level in a recovery tank does not exceed 80%.

Self-contained recovery devices, for use with small appliances containing CFCs, HCFCs, HFCs, and HFOs, must be able to recover 90% of the refrigerant when the compressor is operating or achieve a 4-inch vacuum under the conditions of AHRI 740. With an inoperative compressor, the equipment used to recover CFC, HCFC, and HFC refrigerants from small appliances, for the purpose of disposal, must be able to recover 80% of the refrigerant or achieve 4 inches of vacuum.

Refrigerant recovery devices used to recover CFCs, HCFCs, HFCs and HFOs can be either manually closed or automatically closed when disconnected to prevent loss of refrigerant from hoses. When attaching a gauge set to check system pressures, hoses with manual or self-sealing valves on the ends should be used to minimize the release of refrigerant.

When filling a charging cylinder with a regulated refrigerant, the refrigerant that is vented off the top must be recovered.

A vacuum pump made for system dehydration may be used to evacuate a recovery cylinder to be used as a recovery device, however it cannot be used in combination with a pressurized recovery cylinder or vessel for recovery.

Recovery Requirements

Hydrocarbon (HC) R-600a is an approved refrigerant for new household refrigerators, freezers, and combination refrigeration/freezers. Hydrocarbon refrigerants are not approved for retrofit into existing

household refrigerators designed for refrigerants other than hydrocarbons. New SNAP regulations have approved R-290, R-450A, and R-600a refrigerants for newly manufactured household refrigerators.

Recovery Techniques

When servicing a small appliance found to be leaking, it is not mandatory to repair the leak, but it is recommended to do so whenever possible. Before beginning a refrigerant recovery procedure, it is always necessary to know the type of refrigerant that is in the system to prevent the mixing of refrigerant in the recovery cylinder. It is unacceptable to mix any refrigerants in an appliance or recovery cylinder.

R-744, carbon dioxide CO_2, is a very high-pressure refrigerant and generally does not need to be recovered.

When R-410A or any other refrigerant is recovered from an appliance, it should be recovered into a recovery vessel that is clearly marked as to its content, to ensure that mixing of refrigerants does not occur, Figure 14.

Figure 14.
Recovery Tank.

Comparisons to a pressure-temperature chart are only valid to check for contaminated refrigerant if both the pressure and temperature of the refrigerant are stable and known. When checking for non-condensables inside a recovery cylinder containing one refrigerant, the technician must allow the temperature of the cylinder to stabilize to room temperature before taking a pressure reading for comparison using the pressure-temperature (PT) chart.

If a technician suspects that the refrigerant in a recovery cylinder to be contaminated, a pressure measurement should be taken and compared to a pressure-temperature chart. If the pressure-temperature relationship does not match, the refrigerant should be turned in to a reclamation facility. Some reclamation facilities may refuse to accept mixed refrigerants or charge extra for processing or destroying it.

The service aperture or process stub typically used on small appliances is a straight piece of tubing that is entered using a piercing access valve to add or remove refrigerant from a small appliance. When installing any type of access fitting onto a sealed system the fitting should be leak tested before proceeding with recovery. Because solderless type piercing valves tend to leak over time, they should not remain on the refrigeration systems after completion of the repair.

A passive, system-dependent, recovery process for small appliances can capture refrigerant in a non-pressurized container. When using passive recovery devices on small appliances with non-operating compressors it is essential to take measures to help release trapped, regulated refrigerant from the compressor oil during refrigerant recovery. It is helpful to heat (with a heat gun) and strike the compressor with a rubber mallet when recovering refrigerants into a non-pressurized container from a small appliance, such as a household refrigerator, with an inoperative compressor. Doing so will help free refrigerant that might be trapped under the oil in the compressor crankcase.

When using the system-dependent (passive) recovery process and the compressor does not run, both the high and low side of the system must be accessed for refrigerant recovery.

Both low and high-side access valves should be installed when recovering refrigerant from a household refrigerator or any other appliance to improve the speed of recovery. To ensure that all CFC, HCFC, or HFC refrigerant has been removed from a frost-free refrigerator, the defrost heater can be energized to increase the refrigerant's temperature and vaporize any liquid. An appliance with an operating compressor and completely restricted capillary tube or metering device requires only one access valve on

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the high side of the system. If the pressure in an appliance is 0 psig after installing and opening a piercing access valve, the recovery procedure cannot begin.

When using a self-contained (active) recovery device and there is a large volume of non-condensable gas in the refrigerant being recovered, excessive pressure can occur on the high side of the recovery machine and in the recovery cylinder. Non-condensable gas in the refrigerant can be checked for by using a pressure-temperature (PT) chart. The approximate pressure of an R-134a cylinder that does not contain any non-condensable impurities and is stored in a room at 80°F is equal to a pressure of 87 psig. As long as there is liquid refrigerant present and no impurities in the cylinder, the pressure and temperature should match what is shown on the PT chart.

During system recovery and/or repair, a pungent odor detected from the refrigerant normally indicates a compressor burn-out. The technician should watch for signs of oil contamination when recovering refrigerant from a system that experienced a compressor burn-out as the system will have to be flushed if contaminants are present. When nitrogen is used to pressurize or blow debris out of the system, the nitrogen may be vented to the ambient air.

Safety

Household appliances charged with hydrocarbon (HC) refrigerants require permanent safety markings on or near any evaporators, exposed refrigerant tubing, the machine compartment, and exterior of the refrigerator.

When recovering refrigerants, safe work practices include wearing safety eyewear and butyl-lined gloves when connecting/disconnecting hoses. When using nitrogen to service, leak test, or purge a refrigeration system, the nitrogen tank should be equipped with a regulator.

Because refrigerants are heavier than air and displace oxygen, a very large refrigerant leak can cause suffocation. If a large leak of refrigerant occurs, such as from a filled cylinder in an enclosed area, and no self-contained breathing apparatus is available, vacate and naturally ventilate the area. At high temperatures, CFC and HCFC refrigerants can decompose into phosgene gas, hydrochloric, and/or hydrofluoric acids.